```
 GNU nano 7.2
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t pid = fork();

    if (pid < 0) {
        perror("fork");
        return 1;
    }

    if (pid == 0) {
        printf("Background child started: PID %d\n", getpid());
        sleep(5);
        printf("Background child completed\n");
        _exit(0);
    }

    printf("Parent continues immediately. Child PID: %d\n", pid);
    printf("Use ps to check the background process.\n");

    return 0;
}
```

```c
#include <stdio.h>
#include <unistd.h>
#include <sys/wait.h>
#include <stdlib.h>

int main() {
    pid_t jobs[3];
    int status;

    for (int i = 0; i < 3; i++) {
        jobs[i] = fork();

        if (jobs[i] < 0) {
            perror("fork");
            return 1;
        }

        if (jobs[i] == 0) {
            printf("Job %d running, PID: %d\n",
                   i + 1, getpid());
            sleep(i + 1);
            _exit(i);
        }
    }

    printf("\nJob Table:\n");

    for (int i = 0; i < 3; i++) {
        printf("Job %d | PID %d | Running\n",
               i + 1, jobs[i]);
    }

    for (int i = 0; i < 3; i++) {
        if (waitpid(jobs[i], &status, 0) > 0) {
            printf("Job PID %d completed; exit status %d\n",
                   jobs[i], WIFEXITED(status)
                   ? WEXITSTATUS(status) : -1);
```

```
                ? WEXITSTATUS(status) : -1);
        }
    }

    printf("All jobs completed and removed from tracking.\n");

    return 0;
}

^G Help          ^O Write Out     ^W Where Is      ^K Cut
^X Exit          ^R Read File     ^\ Replace       ^U Paste
```

```
sravya@Sravya1955:~/os$ ./q16a
-bash: ./q16a: No such file or directory
sravya@Sravya1955:~/os$ nano q16a.c
sravya@Sravya1955:~/os$ gcc q16a.c -o q16a
sravya@Sravya1955:~/os$ ./q16a
Parent continues immediately. Child PID: 8541
Use ps to check the background process.
sravya@Sravya1955:~/os$ Background child started: PID 8541
Background child completed

sravya@Sravya1955:~/os$ nano q16b.c
sravya@Sravya1955:~/os$ gcc q16b.c -o q16b
sravya@Sravya1955:~/os$ ./q16b
Job 1 running, PID: 8811

Job 2 running, PID: 8812
Job 3 running, PID: 8813
Job Table:
Job 1 | PID 8811 | Running
Job 2 | PID 8812 | Running
Job 3 | PID 8813 | Running
Job PID 8811 completed; exit status 0
Job PID 8812 completed; exit status 1
Job PID 8813 completed; exit status 2
All jobs completed and removed from tracking.
sravya@Sravya1955:~/os$
```